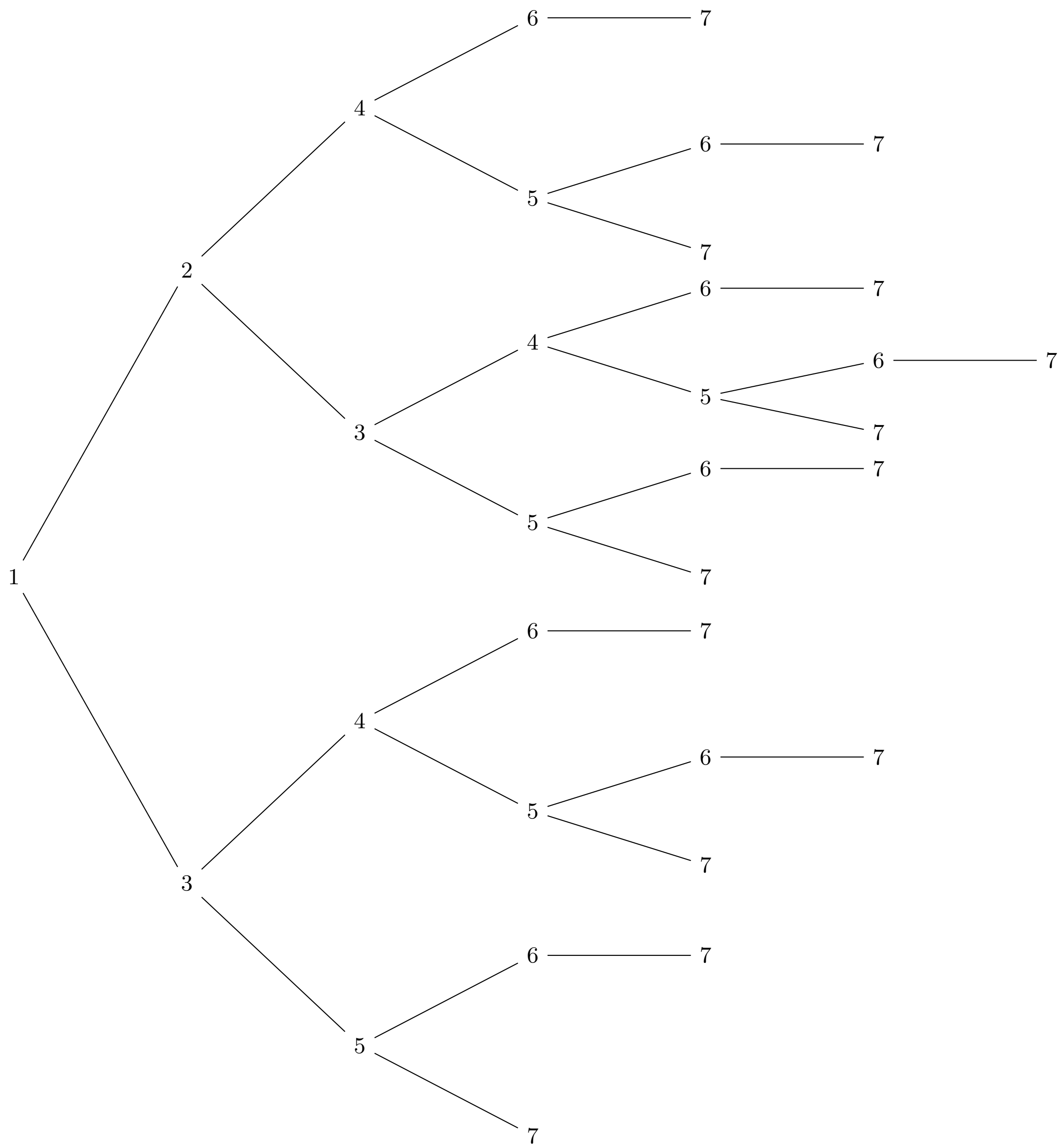
2020 IMC

Q.11
Draw a tree diagram



Q.15
Pythagoras' theorem

Consider the middle height, we have $4h = 3b$.
Consider the right-angled triangle on the left, we have

$$\begin{aligned}
(4h)^2 + (4b)^2 &= 2.4^2 \\
h^2 + b^2 &= \frac{24 \times 24}{1600} = \frac{3 \times 12}{100} \\
h^2 + \frac{16}{9}h^2 &= 0.36 \\
\frac{25}{9}h^2 &= 0.36 \\
h &= \frac{0.6 \times 3}{5} \\
h &= 0.36
\end{aligned}$$

So, we have $b = 0.48$.
Hence, $4b + 3h = 1.92 + 1.08 = 3$.